

UA Field Intel — Business & Functional Requirements Document (BRD/FRD)

Project Name: UA Field Intel **Document Type:** Business & Functional Requirements Document **Prepared For:** Agri-Retail Business Leadership, Product & Engineering Team **Version:** 1.0 (Draft) **Date:** June 12, 2026

1. Document Purpose

This document defines the business problem, objectives, scope, functional and non-functional requirements, data model, and technical approach for **UA Field Intel** — a field data capture and analytics platform for Agricultural Specialist Sales Representatives (ASRs). It is intended to guide the product, design, and engineering teams through discovery, design, and build phases.

2. Background

The company operates an agri-retail network of approximately **100 stores**, each supported by one or more Agricultural Specialist Sales Representatives. These representatives visit farmers in their region to:

- Collect farmer and farm data
- Build long-term relationships
- Promote products relevant to the farmer's crop and season
- Diagnose and recommend solutions for crop-related problems
- Drive footfall, outreach, and ultimately sales conversion at the respective store

2.1 Current State Problem

- Visit data is captured informally (notebooks, memory, personal phones) and remains with the individual representative.
 - No structured record exists of farmer interactions, recommendations made, or problems reported.
 - Lead-to-conversion is not tracked — there is no link between a farmer visit and a subsequent purchase at the store.
 - Regional managers and central teams have no visibility into ground-level activity, farmer needs, or regional trends.
 - Promotion and outreach strategies are not data-driven; they rely on anecdotal feedback.
 - Data, once lost, cannot be recovered — repeat visits to the same farmer may duplicate effort or miss critical history.
-

3. Proposed Solution Overview

UA Field Intel is a mobile-first, low-typing, selection-driven web application (Progressive Web App) that allows Sales Representatives to:

- Log a farmer visit on the spot, with **GPS location confirmation**
- Capture structured farmer, farm, crop, and commercial data primarily via **dropdowns, chips, toggles, and sliders**
- Attach **photos** and **voice notes** as supporting evidence/context
- Save data **offline** and **auto-sync to the cloud** when connectivity is available

This data feeds a **central repository**, which Regional Managers and Central Teams can access through **dashboards and visualizations** to:

- Track lead-to-conversion across the funnel
 - Segment farmers by crop, soil, region, problem type, etc.
 - Identify trends and design targeted promotion/outreach campaigns
 - Monitor representative activity and productivity
-

4. Objectives

1. Digitize 100% of farmer field visits through a simple, fast-entry mobile interface.
 2. Capture accurate, geo-tagged farmer and farm profiles.
 3. Minimize free-text entry — maximize use of pre-defined selections to ensure data consistency and analytics-readiness.
 4. Enable rich context capture via photos and voice notes.
 5. Build a growing, de-duplicated farmer database usable across visits and representatives.
 6. Establish visit → recommendation → follow-up → purchase traceability (lead-to-conversion tracking).
 7. Provide Regional and Central teams with real-time dashboards for decision-making.
 8. Support offline-first operation for rural/low-connectivity areas.
-

5. Scope

5.1 In Scope (Phase 1 — MVP)

- Mobile-responsive web app (PWA) for Sales Representatives
- Visit/Farmer data entry form (per data dictionary in Section 11)

- GPS location capture and confirmation
- Photo capture (multiple) and voice note recording, attached to a visit record
- Offline data capture with auto-sync
- Farmer search/lookup (avoid duplicate entries)
- Central cloud data repository
- Role-based web dashboard for Regional Managers and Central Team (visit volumes, farmer segmentation, basic funnel view)
- Admin module for master data management (crops, soil types, seasons, etc.) and user/store management

5.2 Out of Scope (Phase 1 — Future Phases)

- Native Android/iOS apps (PWA approach covers mobile usage initially)
- Integration with POS/inventory/ERP systems for automated conversion tracking
- AI/ML-based crop or product recommendation engine
- SMS/WhatsApp-based farmer communication
- Multi-language voice transcription / NLP on voice notes
- Loan/finance partner integrations

6. Stakeholders & User Roles

Role	Description
Sales Representative (ASR)	Field user who logs farmer visits, captures data, photos, and voice notes.
Store Manager	Oversees ASRs attached to a store; views store-level visit and farmer data.

Role	Description
Regional Manager	Oversees multiple stores within a region; views regional dashboards and trends.
Central/Analytics Team	HQ team analyzing pan-organization data for strategy, promotions, and reporting.
System Administrator	Manages users, stores, regions, roles, and master/reference data.

6.1 Roles & Permissions Matrix

Capability	ASR	Store Manager	Regional Manager	Central Team	Admin
Create/Edit own visit entries	✓	✓	✗	✗	✗
View own store's visits	✗	✓	✓	✓	✓
View region-wide data	✗	✗	✓	✓	✓
View organization-wide data	✗	✗	✗	✓	✓
Export data	✗	✓	✓	✓	✓
Manage master data (crops, soil, etc.)	✗	✗	✗	✗	✓
Manage users/stores/roles	✗	✗	✗	✗	✓
Update lead/visit status (follow-up)	✓	✓	✓	✗	✗

7. Functional Requirements

7.1 Authentication & User Management

- FR-1.1: Users log in via mobile number/employee ID + OTP or password.
- FR-1.2: Each user is mapped to a Store, Region, and Role.
- FR-1.3: Session persists for offline use; re-authentication required periodically for security.

7.2 Farmer Visit Data Capture (Core Module)

- FR-2.1: ASR can start a **New Visit Entry** or select an **Existing Farmer** (via search by mobile number, name, or village) to log a repeat visit.
- FR-2.2: On starting a new entry, the app captures the **device GPS coordinates** and displays them on a map for the ASR to **confirm or adjust**.
- FR-2.3: The data entry form is presented as a **multi-step wizard** grouped into logical sections (see Section 12).
- FR-2.4: All fields listed in Section 11 are captured, with the input method (dropdown, multi-select, slider, toggle, date picker, numeric, text) as specified — free text is restricted to name, father's name, and a small set of "other/remarks" fields.
- FR-2.5: ASR can attach **multiple photographs** (e.g., farm, crop, problem area, farmer with product).
- FR-2.6: ASR can record and attach **one or more voice notes** (e.g., farmer's problem description, recommendations given).
- FR-2.7: Entry can be **saved as draft** locally if incomplete.
- FR-2.8: On save, the system auto-stamps **Created At** (timestamp) and **Created By** (logged-in ASR), and **Store ID**.

- FR-2.9: If offline, the entry is queued locally and **auto-synced** to the cloud once connectivity is restored, with a visible sync status indicator.

7.3 Farmer Profile & Visit History

- FR-3.1: System maintains a unique Farmer Profile (keyed by mobile number, with name/village as secondary match) consolidating all visits over time.
- FR-3.2: ASR can view a farmer's **visit history** (past crops, problems, recommendations, purchases) before/during a new visit.
- FR-3.3: System flags potential duplicate farmer records for review (same mobile/name+village combination).

7.4 Lead & Follow-Up Tracking

- FR-4.1: Each visit can be tagged with a **Lead/Engagement Status**: e.g., *New, Contacted, Recommendation Given, Follow-up Scheduled, Converted, Lost*.
- FR-4.2: ASR/Store Manager can set a **follow-up date** and receive a reminder/notification.
- FR-4.3: The "Mandi/Last Purchase Shop" field allows linking the farmer's most recent purchase activity to the visit record, enabling a basic conversion view.
- FR-4.4: Status updates are logged with timestamp and user for audit/history.

7.5 Dashboards & Reporting (Regional Manager / Central Team)

- FR-5.1: **Visit Activity Dashboard** — visits logged by day/week/month, filterable by Store, Region, ASR.
- FR-5.2: **Farmer Segmentation Views** — by crop, season, soil type, water source, land size bands, region.
- FR-5.3: **Problem Heatmap** — most commonly reported problems by crop/region/season.

- FR-5.4: **Geo Map View** — visit locations plotted on a map (using captured GPS) for territory coverage analysis.
- FR-5.5: **Lead Funnel View** — counts/percentages by lead status (New → Contacted → Converted).
- FR-5.6: **Media Gallery** — view photos/voice notes attached to specific visits.
- FR-5.7: Export data to Excel/CSV for offline analysis.

7.6 Admin Module

- FR-6.1: Manage **master/reference data** for all dropdown lists (crops, seasons, soil types, water sources, fertilizers, sprays, problems, medicines, animals, machines, mandis, etc.).
- FR-6.2: Manage **Stores, Regions, and Users** (create/deactivate ASRs, assign roles).
- FR-6.3: Manage **geography hierarchy** (State → District → Tehsil → Block → Village → Pincode) for cascading dropdowns.

7.7 Notifications (Phase 1 — basic)

- FR-7.1: In-app notification/reminder for scheduled follow-ups.
- FR-7.2: Sync success/failure notification to ASR.

8. Non-Functional Requirements

Category	Requirement
Usability	Minimal typing; selection-based UI (dropdowns, chips, sliders, toggles); large touch targets suitable for outdoor/field use; support for regional language UI labels (Phase 1: Hindi + English toggle).

Category	Requirement
Offline Support	App must function fully offline for data entry; local storage queue with automatic background sync; clear sync status indicator.
Performance	Form screens load in under 2 seconds on 3G/4G; sync optimized for low-bandwidth (compressed media uploads).
Scalability	Must support ~100 stores, 200–500+ ASR users initially, scalable to 1000+ users and millions of visit records.
Security	Role-based access control; encrypted data in transit (HTTPS) and at rest; audit trail for data edits.
Data Privacy	Farmer personal data (name, mobile, location) handled per applicable data protection norms; consent capture recommended at first visit.
Device Compatibility	Optimized for Android Chrome (primary field device); responsive design for tablets/desktop (manager dashboards).
Availability	Cloud backend targeted at 99.5%+ uptime; field app usable independent of backend availability.
Auditability	All records retain Created At, Created By, and Last Modified metadata.

9. Data Capture Workflow

1. ASR logs into the app (online or with cached session).
2. ASR selects **New Visit** or searches for an **existing farmer**.
3. App captures GPS coordinates; ASR confirms/adjusts location on map.

4. ASR proceeds through the wizard form, selecting values for farmer, land, crop, input usage, problem, and commercial sections.
 5. ASR captures photo(s) and/or records voice note(s).
 6. ASR sets a Lead/Engagement Status and optional follow-up date.
 7. ASR saves the entry — stored locally if offline, or pushed directly to cloud if online.
 8. Background sync pushes queued entries to the central repository when connectivity is available.
 9. Data becomes visible to Store Manager, Regional Manager, and Central Team dashboards in near real-time post-sync.
 10. Central Team uses aggregated data to plan region/crop-specific promotions; follow-up tasks may be routed back to ASRs.
-

10. UI/UX Design Principles

- **Wizard-style form** broken into sections (suggested order):
 1. Farmer Identity (Name, Father's Name, Mobile)
 2. Location (Village, Post, Block, Tehsil, District, State, Pincode + GPS confirm)
 3. Land & Soil (Total Land, Irrigated Land, Soil Type, Water Source)
 4. Crop & Season (Crop, Season, Sowing Date, Expected Harvest)
 5. Inputs Used (Seed, Fertilizer, Spray)
 6. Problem & Advisory (Problem, Medicine/Recommendation)
 7. Commercial (Budget, Loan, Mandi, Last Purchase Shop)
 8. Assets (Animals, Tractor, Machine)
 9. Media (Photos, Voice Note)
 10. Status & Follow-up
- Each section shows **progress indicator** (e.g., "Step 4 of 10").

- Wherever possible: **single-tap chip selection, multi-select tag clouds, sliders for numeric ranges** (e.g., land size), **toggle switches for Yes/No** (e.g., Loan, Tractor ownership).
- Free text limited to: Farmer Name, Father's Name, and optional "Other/Remarks" fields.
- Cascading location dropdowns (State → District → Tehsil → Block → Village) to speed entry and ensure standardization.
- Persistent **offline/sync status badge** visible at all times.
- "Save & New" action to quickly start the next visit.

11. Data Dictionary (Core Fields)

Field	Description	Input Type	Mandatory	Notes / Suggested Options
Farmer Name	Full name of farmer	Text	Yes	
Father Name	Father's/Husband's name	Text	No	
Mobile	Contact number	Numeric (10-digit)	Yes	Used as primary de-dup key
Village	Village name	Dropdown (cascading)	Yes	Linked to Block
Post	Post office	Dropdown/Text	No	

Field	Description	Input Type	Mandatory	Notes / Suggested Options
Block	Administrative block	Dropdown (cascading)	Yes	Linked to Tehsil
Tehsil	Tehsil/Taluka	Dropdown (cascading)	Yes	Linked to District
District	District	Dropdown (cascading)	Yes	Linked to State
State	State	Dropdown	Yes	
Pincode	Postal code	Numeric / auto- fill	No	Auto-suggest from Village
Total Land	Total landholding	Numeric/Slider	Yes	Units: Acres/Hectares (configurable)
Irrigated Land	Irrigated portion of land	Numeric/Slider	No	≤ Total Land
Soil	Soil type	Dropdown	No	Sandy, Loamy, Clayey, Black, Red, Alluvial, Saline, Other
Water Source	Primary irrigation source	Dropdown (multi-select)	No	Borewell, Canal, River, Pond/Tank,

Field	Description	Input Type	Mandatory	Notes / Suggested Options
				Rainfed, Drip/Sprinkler
Crop	Current/planned crop	Dropdown (multi-select)	Yes	Master crop list
Season	Cropping season	Dropdown	Yes	Kharif, Rabi, Zaid
Sowing Date	Date of sowing	Date Picker	No	
Harvest	Expected harvest date/yield	Date Picker / Numeric	No	
Seed	Seed brand/variety used	Dropdown	No	Master seed list (brand + variety)
Fertilizer	Fertilizers used	Dropdown (multi-select)	No	Master fertilizer list
Spray	Pesticide/herbicide/fungicide used	Dropdown (multi-select)	No	Master spray list
Problem	Issue(s) reported by farmer	Dropdown (multi-select, categorized)	No	Pest, Disease, Nutrient Deficiency, Water Stress, Weather Damage,

Field	Description	Input Type	Mandatory	Notes / Suggested Options
				Market/Price, Other
Medicine	Recommended treatment/product	Dropdown	No	Linked to Problem category
Budget	Farmer's stated budget for inputs	Numeric / Range Selector	No	Range bands (e.g., <5k, 5-10k, 10-25k, 25k+)
Loan	Whether farmer has taken a loan	Toggle (Yes/No) + Dropdown	No	If Yes: Bank, Cooperative, Govt Scheme, Private Lender
Animals	Livestock owned	Multi-select + count	No	Cow, Buffalo, Goat, Sheep, Poultry, Other
Tractor	Tractor ownership	Toggle (Yes/No) + Dropdown	No	If Yes: Brand/Model (optional)
Machine	Farm equipment owned	Multi-select	No	Rotavator, Thresher, Sprayer, Harvester, Other
Mandi	Nearest/preferred market	Dropdown	No	Master mandi list

Field	Description	Input Type	Mandatory	Notes / Suggested Options
Last Purchase Shop	Store/shop of last purchase	Dropdown	No	Master store list; supports conversion tracking
Photos	Field/crop/problem photos	Media Upload (multi)	No	Stored in cloud object storage
Voice Note	Audio recording	Media Upload (audio)	No	Stored in cloud object storage
Lead Status	Engagement/conversion status	Dropdown	Yes	New, Contacted, Recommendation Given, Follow-up Scheduled, Converted, Lost
Follow-up Date	Next scheduled contact	Date Picker	No	
GPS Coordinates	Visit location	Auto-captured (Lat/Long)	Yes	ASR confirms/adjusts on map
Created At	Record creation timestamp	Auto	Yes	System-generated

Field	Description	Input Type	Mandatory	Notes / Suggested Options
Created By	ASR who created the record	Auto	Yes	From logged-in session
Note: All "Master" lists (crops, soils, fertilizers, sprays, problems, medicines, mandis, stores, geography hierarchy) are maintained centrally via the Admin Module and should be versioned/extendable without app redeployment.				

12. High-Level Technical Architecture

Layer	Recommended Approach
Frontend	Progressive Web App (PWA) — responsive, installable on mobile home screen, works offline via service workers
Local Storage	IndexedDB / local storage for offline form data and media queueing
Backend/API	REST or GraphQL API layer for CRUD operations, sync, and authentication
Database	Relational DB (e.g., PostgreSQL) for structured farmer/visit data
Media Storage	Cloud object storage (e.g., S3-compatible) for photos and voice notes
Authentication	Token-based (JWT/OAuth2) with role-based access control
Analytics/BI Layer	Dashboarding tool (e.g., Metabase, Power BI, or custom React dashboard) connected to the central DB/warehouse

Layer	Recommended Approach
Maps/Geo	Maps API for GPS confirmation and geo-visualization (e.g., Google Maps/OpenStreetMap)
Hosting	Cloud platform (AWS/Azure/GCP) with auto-scaling for API and DB layers

13. Reporting & Analytics KPIs

- Number of visits logged per ASR / Store / Region / Day-Week-Month
- Active vs. inactive farmer database growth over time
- Crop-wise, soil-wise, region-wise farmer distribution
- Top reported problems by crop and season
- Lead funnel conversion rates (New → Converted) by Store/Region/ASR
- Average time from visit to follow-up to conversion
- Coverage map: visited villages/territories vs. total addressable area
- Data completeness score per visit entry (% of fields filled)

14. Assumptions & Constraints

1. ASRs are equipped with Android smartphones with intermittent internet connectivity.
2. Farmer consent for data collection is obtained verbally/in-app at first visit (consent capture to be added if required by policy).
3. Master/reference data (crops, products, geography) will be provided/maintained centrally before go-live.

- 4. GPS accuracy depends on device hardware and field conditions; ASR confirmation step mitigates inaccuracies.
 - 5. Phase 1 focuses on data capture and visibility; deep system integrations (POS/ERP) are future-phase.
-

15. Success Metrics

Metric	Target (Indicative)
% of farmer visits logged digitally	≥ 90% within 6 months of rollout
Average data completeness per entry	≥ 80% of fields populated
Time to sync after connectivity restored	< 5 minutes for queued entries
Reduction in "lost lead" instances	Measurable decline quarter-over-quarter
Regional dashboard adoption (Managers logging in weekly)	≥ 80% of Regional Managers
Lead-to-conversion visibility	100% of visits taggable to a lead status

16. Implementation Roadmap (Indicative Phasing)

Phase	Scope	Indicative Duration
Phase 1 — MVP	Core data capture form (all fields), GPS confirmation, photo/voice attachment, offline + sync, basic auth, central DB, basic dashboard (visit counts, segmentation)	8-12 weeks
Phase 2	Lead/follow-up workflow, farmer profile & visit history, duplicate detection, advanced dashboards (funnel, heatmaps, geo-map)	4-6 weeks
Phase 3	Admin module for master data/geography management, export tools, regional language UI	4-6 weeks
Phase 4 (Future)	POS/ERP integration for conversion tracking, SMS/WhatsApp follow-up, AI-based recommendations, native app	TBD

17. Risks & Mitigation

Risk	Mitigation
Poor/no connectivity in rural areas	Offline-first design with local queue and background sync
Low user adoption due to complex forms	Selection-driven UI, minimal typing, wizard flow, field training

Risk	Mitigation
Inconsistent/poor-quality data entry	Mandatory fields, dropdown-driven inputs, validation rules
Duplicate farmer records	Mobile-number-based de-dup logic with manual merge review
GPS inaccuracy	ASR confirmation/adjustment step before save
Data privacy concerns	Role-based access, encryption, consent capture, audit logs
Media storage costs (photos/voice) at scale	Client-side compression before upload; storage lifecycle policies

18. Glossary

Term	Meaning
ASR	Agricultural Specialist Sales Representative
PWA	Progressive Web App
Lead Status	Stage of farmer engagement in the conversion funnel
Master Data	Centrally maintained reference lists (crops, soils, products, geography, etc.)
Mandi	Local agricultural market/marketplace
De-dup	De-duplication — identifying and merging duplicate farmer records

19. Open Items for Stakeholder Discussion

1. Confirm preferred unit of measurement for land (Acres / Hectares / Bigha — region-dependent).
2. Confirm regional language priorities for UI localization.
3. Confirm data retention and farmer consent/privacy requirements applicable to the organization.
4. Confirm whether store-level POS data can be made available (even via manual export) for conversion tracking in later phases.
5. Confirm device standardization (Android version baseline) across the 100-store network.

End of Document